BY ORDER OF THE COMMANDER
AIR FORCE SUSTAINMENT CENTER

AIR FORCE MATERIEL COMMAND
INSTRUCTION 21-100 Volume 1, Volume 2, Volume 3

AIR FORCE SUSTAINMENT CENTER
Volume 1, Volume 2, Volume 3 Supplement, ADDENDUM A

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Maintenance

AFSC SOFTWARE DIRECTORATE MANAGEMENT

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This addendum implements and extends Air Force Material Command Instruction (AFMCI) 21-100, Air Force Sustainment Center (AFSC) Supplement Volume 1, *Depot Maintenance Principles,* Volume 2 *Depot Maintenance Production and* Volume 3 *Depot Maintenance Production Support*. This publication provides the minimum basic logistics, engineering, and general Department of the Air Force (DAF) program requirements for the AFSC Software Directorate (AFSC/SW) and for any AFSC/SW organization supporting either United States Air Force (USAF) or United States Space Force (USSF). AFSC/SW is organized as an AFSC Non-Standard Organization (NSO) in accordance with (IAW) AFI 38-101, AFMC SUP, *Manpower and Organization* and requires supplemental guidance. This publication applies to all AFSC/SW personnel. The AFSC NSOs require approval to deviate (applicability, variance, and/or differences in organizational placement of responsibilities/processes) from AFMCI 21-100 AFSC Sup Vol 1-3. The AFSC/SW has been authorized by AFSC Logistics Directorate (AFSC/LG) to deviate from the processes and procedures in AFMCI 21-100 Vol 1-3 and AFMCI 21-100 AFSC Supplement Vol 1-3. AFSC/SW personnel will comply with this Addendum only, not AFMCI 21-100 volumes and AFMCI 21-100 AFSC Supplement volumes. This addendum scopes AFSC/SW requirements in meeting the intent of the parent AFMCI and the AFSC Supplement based on its unique mission. This addendum aligns under publication series 21-Maintenance IAW DAFMAN 90-161 *Publishing Processes and Procedures* to maintain the line of authority and alignment to AFSC sustainment processes,

programs, and parent documents. Organizations that implement processes that were not identified as applicable in this Addendum or gain those functions will follow the guidance in AFMCI 21-100 AFSC Sup volumes or submit a deviation in the form of a waiver request. All contractor requirements contained within this Addendum must be contained within the contract/grant/agreement to be enforceable. Ensure that all records created because of processes prescribed in this publication are maintained in accordance with (IAW) AFI 33-322, *Records Management and Information Governance Program,* and disposed of IAW Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the form DAF 847, *Recommendation for Change of Publication*. Route the DAF 847 from the field through the appropriate functional chain of command. The waiver authority for this publication is AFSC/LZ. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force. Units developing separate publications based on requirements set forth in this publication will follow guidance in DAFI 90-160, *Publications and Forms Management.*

SUMMARY OF CHANGES

This publication has been created to meet the intent of AFMCI 21-100 AFSC Sup Vol 1-3 requirements as applicable to the AFSC/SW; review this addendum in its entirety.

Chapter 1

MANAGEMENT PHILOSOPHY AND APPLICABILITY

1.1. Overview. This Addendum was constructed to establish guidance for AFSC/SW. AFSC/LG will provide the minimum business, logistics, engineering, and general AF program requirements for AFSC/SW and may use additional management processes not specifically prohibited by this instruction, any technical order, or other applicable instruction.

1.2. AFSC/SW will: Provide directorate level guidance and staff coordination for all software organizations developing software and hardware or providing technical services.

1.3. Commander/Director's Intent. Organizational vision, mission and director's intent will be provided in the AFSC/SW Strategic Plan.

1.4. Guidance Development. At a minimum, AFSC/SW will publish specific guidance (e.g., Supplements, Instructions) in the appropriate publication series as required for the programs outlined in this instruction.

1.5. Software Engineering Discipline. Personnel will comply with written guidance when directed IAW AFIs, TOs, agreements, and customer contracts.

1.6. Workplace Communications. Effective software and hardware development requires effective communication. Directorate and group commanders/directors will develop communication plans according to mission requirements.

1.7. Compliance Terminology Definitions. For this instruction, the following definitions apply:

1.7.1. Will, Must - Indicate mandatory requirements.

1.7.2. Should - Indicates a preferred method of accomplishment.

1.7.3. May - Indicates an acceptable or suggested means of accomplishment.

1.8. Waivers. Any waivers to this addendum will be processed IAW higher headquarters directives. Waiver requests will be submitted IAW DAFMAN 90-161.

Chapter 2

MANAGEMENT PROGRAMS

2.1. Strategic Agility. Establish guidance for portfolio management, customer engagement, strategic planning, workload assignment, depot activation, resource management, and partnership initiatives for both private and public sectors. Identify production analytics and capability sourcing, and metrics reporting that enables the software ecosystem framework.

2.2. Work Measurement. Define the labor classification of personnel type (direct, overhead, General and Administrative (G&A)) and the type of work personnel perform (direct, indirect, overhead) to meet the requirements of both external and internal customers.

2.3. Capital Investment Program (CIP). Implement AFSC CIP requirements.

2.4. Production Support. Establish guidance for software and hardware engineering development and sustainment production support activities to include workload control, planning, and documentation requirements.

2.5. Operational Workloading, Planning, and Scheduling Controls. Establish guidance to manage operational workloading, planning and scheduling controls. Define controls and processes applicable to each workload from the point before induction through job closure.

2.6. Labor Entry. Develop guidance to ensure labor entry requirements are a standard means of entering labor transactions with minimal effort from the affected employee. Use Depot Maintenance Accounting and Production System (DMAPS) Time and Attendance (TAA) or other-directed information system to process labor and attendance for Consolidated Sustainment Activity Group-Maintenance (CSAG-M) funded employees.

2.7. Capacity and Utilization Measurement. Implement AFSC requirements for AFMC capacity and utilization measurement as required of all Department of Defense (DoD) depot maintenance activities to quantify organic capacity and capacity utilization providing indicators of the relative size and usage of DoD organic depot maintenance activities.

Chapter 3

QUALITY MANAGEMENT SYSTEM (QMS)

3.1. System Engineering Expectations. Establish engineering expectations in alignment with AFMCI 63-1201, *Integrated Life Cycle Systems Engineering and Technical Management* for implementation by production units to meet organizational expectations and customer requirements.

3.2. Quality Assurance (QA). Establish QA guidance to provide adequate confidence that controls are in place to ensure processes conform to established requirements, standards, and organizational expectations. Process conformance will be IAW all technical, safety, and other applicable publications. Quality processes will include:

3.2.1. Roles and responsibilities of key personnel involved in the QA program.

3.2.2. Surveillance and training plans.

3.2.3. Types of assessments.

3.2.4. Corrective action and preventive action requirements to include root cause.

3.2.5. Standardized documentation and information system.

3.2.6. Data collection, analysis, and reporting.

3.3. Unit Self-Assessment Program (USAP). Establish a USAP led by unit commanders and directors IAW AFI 1-2, *Commander's Responsibilities*. The USAP provides a means for internal assessment of overall organizational health and complements external assessments.

3.3.1. Comply with DAFI 90-302, *Air Force Inspection System* and implement *AFSC USAP, Management Internal Control Toolset (MICT) and Inspector General Evaluation Management System (IGEMS) Business Rules Handbook* for business processes and general AF programs.

3.3.2. Establish internal engineering self-assessments aligned with engineering principles and organizational expectations.

3.4. Corrective Action and Continuous Process Improvement (CPI). Establish guidance to institutionalize the use of continuous process improvement and innovation concepts, tools, techniques, and processes into organizational culture.

3.4.1. Promote and use CPI methodologies to fulfill requirements in AFI 1-2, DAFI 90-302, and AFI 38-401, *Continuous Process Improvement (CPI).*

3.4.2. Support root cause and corrective action plan activities for QA and Inspector General (IG) deficiencies as well as USAP self-identified observations.

3.4.3. Support organizational CPI projects to address gaps between current and standard performance measures.

3.5. Art of the Possible (AoP). Implement AoP to meet mission requirements IAW AFSCI 60-101, *Art of the Possible* and AFSCH 60-101, *Art of the Possible Handbook.*

Chapter 4

GENERAL PROGRAMS

4.1. Electrostatic Discharge (ESD) Program. Establish guidance for personnel handling and working on items that meet ESD criteria IAW TO 00-25-234*, General Shop Practice Requirements for the Repair, Maintenance, and Test of Electrical Equipment*. Local ESD directives and technical data may outline stricter controls.

4.2. Equipment Maintenance. Establish guidance for management of Test, Measurement and Diagnostic Equipment (TMDE), Support Equipment (SE), machinery, and shop equipment IAW TO 00-20-14, *Air Force Metrology and Calibration Program,* TO 00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policy and Procedures,* and TO 34-1-3, *Machinery and Shop Equipment.*

4.3. Foreign Object Damage (FOD) Awareness. Establish guidance for FOD Awareness program to include applicability, responsibilities, and training requirements.

4.4. Housekeeping. Establish guidance to maintain safe, clean (including hygiene), organized, and operationally functional work areas (laboratories, software control centers, local information technology (IT) areas, offices, warehouses, etc.) and hardware products with an emphasis on a culture of tidiness and cleanliness to reduce accidents, machine/equipment failures, and improve overall effectiveness of software/hardware products.

4.5. Materiel Control. Establish guidance for the management and control of materiel. Personnel will comply with AFI 23-101, *Materiel Management Policy.*

4.6. Safety. Comply with DAFI 91-202, *The US Air Force Mishap Prevention Program*, DAFMAN 91-203, *Air Force Occupational Safety Fire and Health Standards,* AFI 90-802*, Risk Management* and other applicable guidance to protect personnel, products, and material to maximize capabilities while limiting risks.

4.7. Technical Orders (TO) and Technical Data. Establish guidance for the use, handling, and management of developed source data and Operation and Maintenance (O&M) TOs and technical data.

4.8. Tool and Equipment Program. Establish guidance to effectively control, secure, and account for tools and equipment. Tool and equipment program requirements should be based on the individual program/project mission and risk.

4.9. Training. Establish guidance to identify training for workload requirements, organizational requirements, and general AF programs. Training requirements are directed IAW equivalent AF instructions, manuals, publications, associated supplements, TOs, and technical data etc. Track training in an AFSC/SW approved IS.

Chapter 5

SECURITY PROGRAMS

5.1. Security. Implement a robust security program to encompass Information, Industrial, and Personnel security. Develop a written security instruction IAW DoDM 5200.01_DAFMAN 16-1404 Volume 1, *Information Security Program: Overview, Classification, And Declassification.* The instruction will include Information Security, Personnel Security and Industrial Security Programs.

5.2. Cybersecurity. Create a culture of cyber awareness, discipline, and compliance that focuses on daily cyber hygiene activities which requires continuous attention to mitigate daily threats. Personnel will comply with applicable AFI 17-series guidance, technical data, host base guidance, and local and visiting area guidance and policies.

JENNIFER HAMMERSTEDT
Lieutenant General, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DoDM 5200.01_DAFMAN 16-1404 Volume 1, *Information Security Program: Overview, Classification, And Declassification,* 5 April 2022

DAFI 90-160*, Publications and Forms Management,* 13 April 2022

DAFI 90-302, *The Inspection System of the Department of the Air Force*, 15 March 2023

DAFI 91-202, *The US Air Force Mishap Prevention Program,* 11 March 2020

AFI 1-2, *Commander's Responsibilities,* 7 May 2014

AFI 23-101, AFMC Sup, *Materiel Management Policy,* 9 December 2021

AFI 33-322, *Records Management and Information Governance Program*, 28 July 2021

AFI 38-101, AFMC SUP, *Manpower and Organization,* 6 July 2021

AFI 90-802*, Risk Management,* 1 April 2019

AFSCI 60-101, *Art of the Possible*, 13 April 2021

AFSCH 60-101 *Art of the Possible Handbook*, 30 August 2023

DAFMAN 90-161, *Publishing Processes and Procedures*, 18 October 2023

DAFMAN 91-203, *Air Force Occupational Safety, Fire and Health Standards*, 25 March 2022

AFMAN 17-1301, *Computer Security*, 12 February 2020

AFMAN 17-1302-O, *Communications Security (COMSEC) Operations,* 9 April 2020

TO 00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures*, 26 September 2022

TO 00-20-14, *Air Force Metrology and Calibration Program*, 28 February 2021

TO 00-25-234*, General Shop Practice Requirements for the Repair, Maintenance, and Test of Electrical Equipment,* 27 October 2021

TO 34-1-3, *Machinery and Shop Equipment,* 23 August 2019

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication,* 15 April 2022

Abbreviations and Acronyms

AFI—Air Force Instruction

AFMC—Air Force Materiel Command

AFMCI—Air Force Materiel Command Instruction

AFRIMS—Air Force Records Information Management System

AFSC—Air Force Sustainment Center

CIP—Capital Investment Program

DMAPS—Depot Maintenance Accounting and Production System

DoD—Department of Defense

ESD—Electrostatic Discharge

FOD—Foreign Object Damage

G&A—General and Administrative

IG—Inspector General

IS—Information System

IT—Information Technology

NSO—Non-Standard Organization

O&M—Operations and Maintenance

OPR—Office of Primary Responsibility

QA—Quality Assurance

QMS—Quality Management System

RDS—Records Disposition Schedule

TAA—Time and Attendance

TMDE—Test, Measurement and Diagnostic Equipment

TO—Technical Order

USAF—United States Air Force

USAP—Unit Self-Assessment Program

USSF—United States Space Force